Logistic Regression

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Imports

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.cm as cm
from sklearn.model_selection import train_test_split , cross_val_score, Stratifi
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score, classification_report, ConfusionMatr
from sklearn import tree
from sklearn.model_selection import train_test_split , cross_val_score, Stratifi
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
import seaborn as sns
from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    classification_report
)
from sklearn.metrics import roc_curve, auc
```

Load Data

In [2]:
```
spaceship_data = pd.read_csv('final_data.csv')
```

split to x and y

In [3]:
```
y = spaceship_data["Transported"].astype(int)
x = spaceship_data.drop(columns=["Transported"])
```

Train-Test Split

In [4]:
```
x_train, x_test, y_train, y_test = train_test_split(
    x, y, test_size=0.2, random_state=42
)
```

In [5]:
```
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
```

בדיקת ערכי ההייפר פרמטרים

In [6]:
```
results = []
best_score = 0
best_params_lr = None
```

In [7]:
```
C_list = [0.01, 0.1, 1, 10, 100]
penalties = ["l2", "l1"]

for C in C_list:
    for pen in penalties:

        # solver must match penalty
        solver = "liblinear" if pen in ["l1", "l2"] else "lbfgs"

        lr_pipe = Pipeline([
            ("scaler", StandardScaler()),
            ("lr", LogisticRegression(
                C=C,
                penalty=pen,
                solver=solver,
                max_iter=5000,
                random_state=42
            ))
        ])

        fold_scores = cross_val_score(
            lr_pipe,
            x_train, y_train,
            cv=cv,
            scoring="accuracy",
            n_jobs=-1
        )

        mean_score = fold_scores.mean()

        results.append({
            "penalty": pen,
            "C": C,
            "cv_mean": mean_score,
            "cv_std": fold_scores.std()
        })

        if mean_score > best_score:
            best_score = mean_score
            best_params_lr = {"penalty": pen, "C": C}
```

In [8]:
```
print("\nBest Logistic Regression params:", best_params_lr)
print(f"Best CV accuracy: {100 * best_score:.2f}%")

results_df = pd.DataFrame(results).sort_values("cv_mean", ascending=False)
display(results_df.head(10).round(4))
```

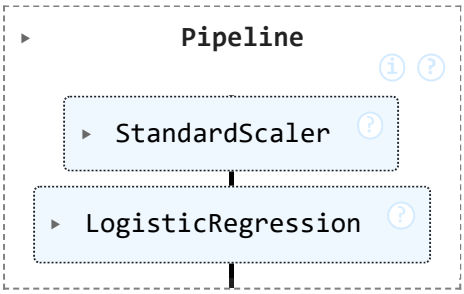
```
Best Logistic Regression params: {'penalty': 'l1', 'C': 10}
Best CV accuracy: 79.07%
```

	penalty	C	cv_mean	cv_std
9	l1	100.00	0.7907	0.0114
8	l2	100.00	0.7907	0.0114
7	l1	10.00	0.7907	0.0114
6	l2	10.00	0.7906	0.0112
4	l2	1.00	0.7904	0.0111
5	l1	1.00	0.7901	0.0111
3	l1	0.10	0.7893	0.0098
2	l2	0.10	0.7887	0.0118
0	l2	0.01	0.7787	0.0105
1	l1	0.01	0.7691	0.0118

best model

In [9]:
```
best_LR = Pipeline([
    ("scaler", StandardScaler()),
    ("lr", LogisticRegression(
        C=best_params_lr["C"],
        penalty=best_params_lr["penalty"],
        solver="liblinear",
        max_iter=5000,
        random_state=42
    ))
])

best_LR.fit(x_train, y_train)
```

Out[9]:

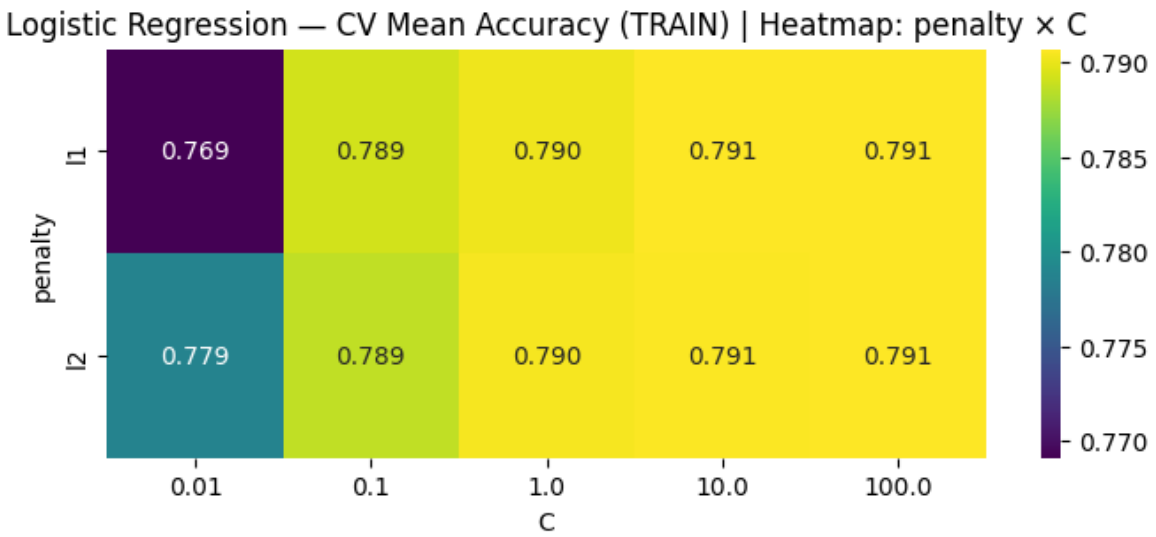


Results

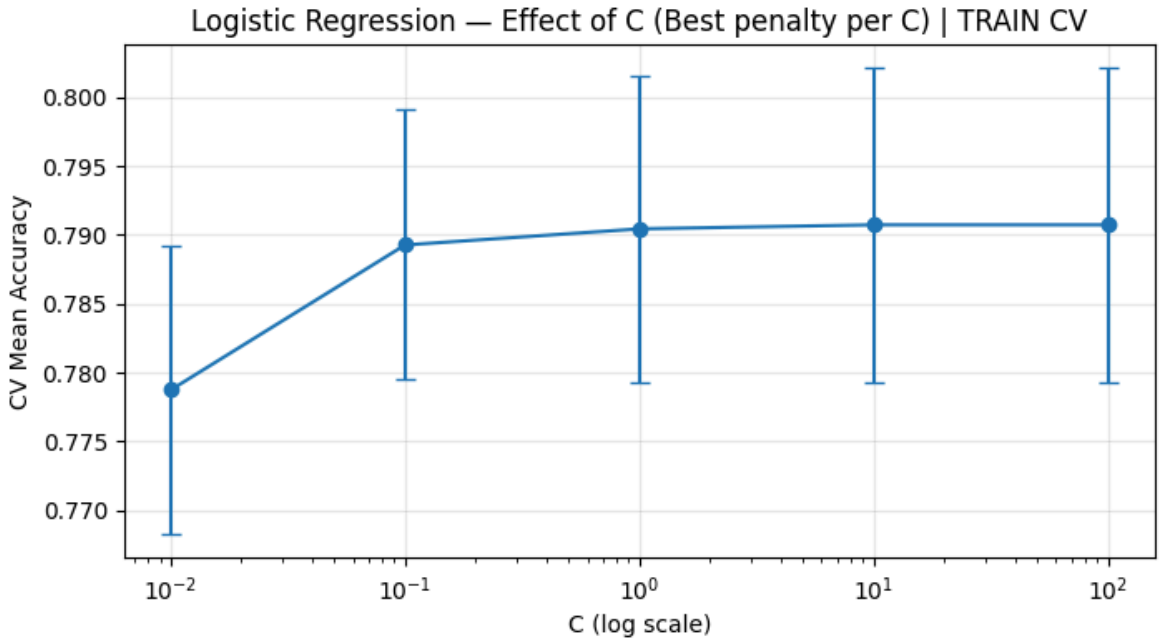
In [10]:
```
y_pred_train = best_LR.predict(x_train)
accuracy_train = accuracy_score(y_train, y_pred_train)

print("="*40)
print("BEST Logistic Regression – TRAIN SET PERFORMANCE")
print("="*40)
print(f"Accuracy : {accuracy_train*100:.2f}%")
print("="*40)
```

```
========================================
BEST Logistic Regression — TRAIN SET PERFORMANCE
========================================
Accuracy : 79.29%
========================================
```

In [11]:
```
hm = results_df.pivot(index="penalty", columns="C", values="cv_mean")

plt.figure(figsize=(8,3))
sns.heatmap(hm, annot=True, fmt=".3f", cmap="viridis")
plt.title("Logistic Regression — CV Mean Accuracy (TRAIN) | Heatmap: penalty × C
plt.xlabel("C")
plt.ylabel("penalty")
plt.show()
```

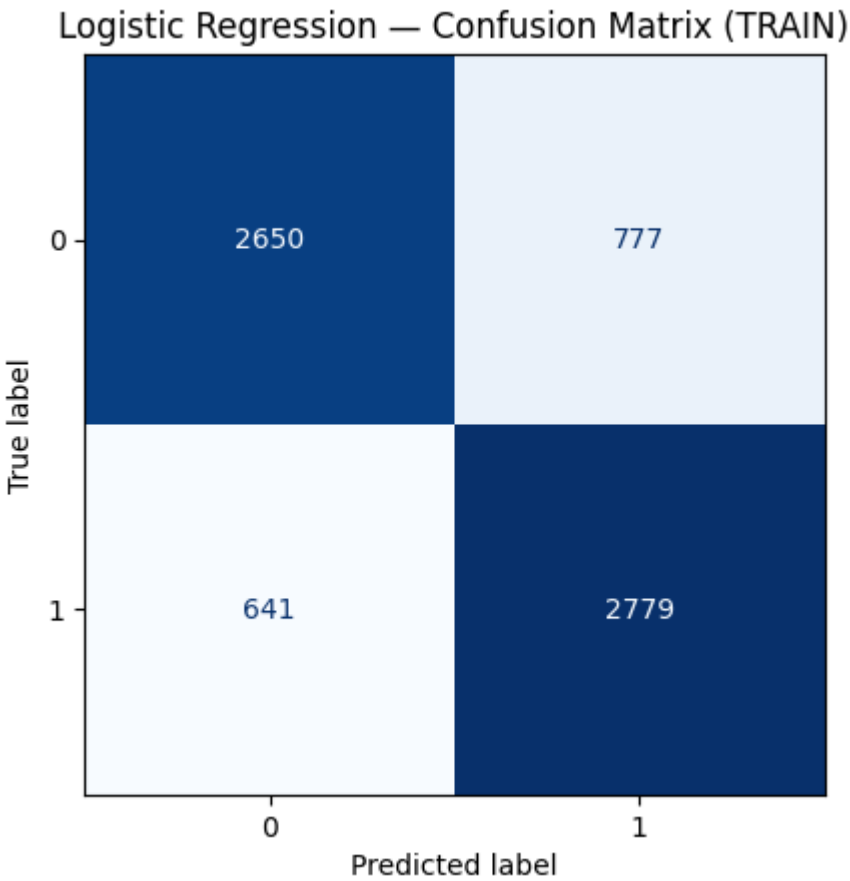


In [12]:
```
plot_c = (
    results_df
    .sort_values(["C", "cv_mean"], ascending=[True, False])
    .groupby("C", as_index=False)
    .head(1)
    .sort_values("C")
)

plt.figure(figsize=(8,4))
plt.errorbar(plot_c["C"], plot_c["cv_mean"], yerr=plot_c["cv_std"], fmt="o-", ca
plt.xscale("log")
plt.title("Logistic Regression — Effect of C (Best penalty per C) | TRAIN CV")
plt.xlabel("C (log scale)")
plt.ylabel("CV Mean Accuracy")
plt.grid(alpha=0.3)
plt.show()
```



In [13]:
```
plt.figure(figsize=(5,5))
ConfusionMatrixDisplay.from_predictions(
    y_train,
    y_pred_train,
    cmap="Blues",
    colorbar=False
)
plt.title("Logistic Regression — Confusion Matrix (TRAIN)")
plt.show()
```

```
<Figure size 500x500 with 0 Axes>
```

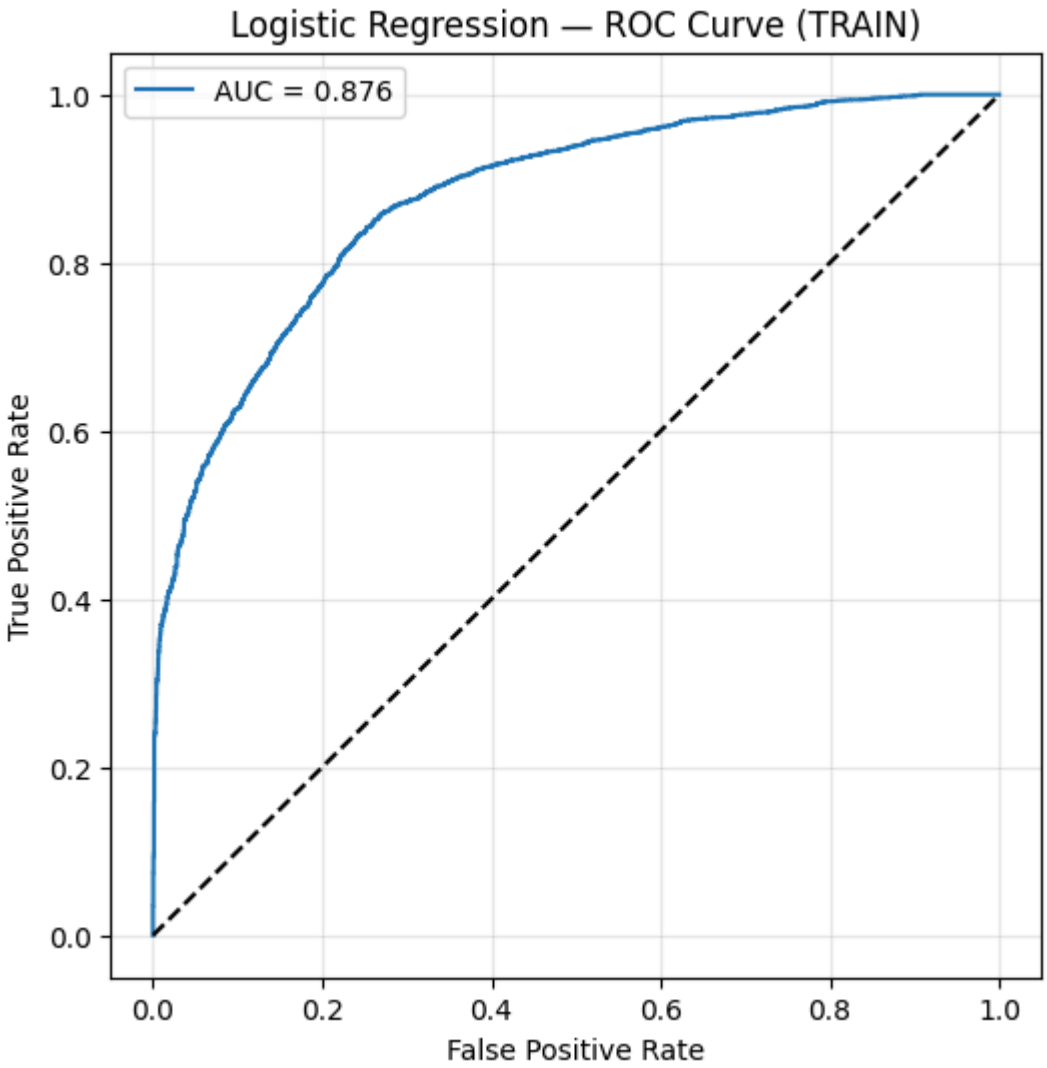


In [14]:
```
acc = accuracy_score(y_train, y_pred_train)
prec = precision_score(y_train, y_pred_train)
rec = recall_score(y_train, y_pred_train)
f1 = f1_score(y_train, y_pred_train)

metrics_df = pd.DataFrame([{
    "accuracy": acc,
    "precision": prec,
    "recall": rec,
    "f1_score": f1
}])

display(metrics_df.round(4))

print("\nClassification Report:")
print(classification_report(y_train, y_pred_train))
```

	accuracy	precision	recall	f1_score
0	0.7929	0.7815	0.8126	0.7967

```
Classification Report:
              precision    recall  f1-score   support

           0       0.81      0.77      0.79      3427
           1       0.78      0.81      0.80      3420

    accuracy                           0.79      6847
   macro avg       0.79      0.79      0.79      6847
weighted avg       0.79      0.79      0.79      6847
```

In [15]:
```
y_prob_train = best_LR.predict_proba(x_train)[:,1]

fpr, tpr, thresholds = roc_curve(y_train, y_prob_train)
roc_auc = auc(fpr, tpr)

plt.figure(figsize=(6,6))
plt.plot(fpr, tpr, label=f"AUC = {roc_auc:.3f}")
plt.plot([0,1],[0,1],'k--')

plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("Logistic Regression – ROC Curve (TRAIN)")
plt.legend()
plt.grid(alpha=0.3)
plt.show()
```


Logistic Regression — ROC Curve (TRAIN)
AUC = 0.876
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0